

CONTRAST-ENHANCED MAMMOGRAPHY AND DIGITAL BREAST TOMOSYNTHESIS FINDINGS OF A BORDERLINE PHYLLODES TUMOR WITH CONTRALATERAL FIBROADENOMA: A CASE REPORT

Anggit Yoso Sulistyowati¹, Lina Choridah², Rahmawati Istiningrum³

¹ Departement of Radiology, RSUD Loekmono Hadi Kudus, Jawa Tengah 59348, Indonesia
² Departement of Radiology, RSUP Dr Sardjito / Faculty of Medicine, Public Health, and Nursing Universitas Gadjah Mada, Yogyakarta 55281, Indonesia
³ Departement of Radiology, RSU Dadi Keluarga Purwokerto, Jawa Tengah 53145, Indonesia



BACKGROUND

Borderline phyllodes tumors are uncommon fibroepithelial breast neoplasms that may overlap radiologically and histologically with fibroadenomas. Contrast-enhanced mammography (CEM) may provide additional information regarding lesion vascularity beyond conventional mammography and digital breast tomosynthesis (DBT).

OBJECTIVES

Primary objective:
To present the CEM and DBT findings of a borderline phyllodes tumor in the left breast and fibroadenoma in the right breast.

Secondary objective(s):
To highlight the potential role of CEM in differentiating similar fibroepithelial breast lesions.

Outcome:
Accurate imaging characterization to guide tissue diagnosis and management planning.

METHODS

- **Design:** Case report
- **Setting:** Tertiary hospital
- **Inclusion criteria:** Woman with palpable breast mass who underwent bilateral 2D mammography, DBT, and CEM
- **Imaging protocol:**
 - 2D mammography (CC, MLO)
 - Digital breast tomosynthesis/DBT (CC, MLO)
 - Contrast-enhance mammo-graphy/CEM (CC, MLO)
- **Reference standard:** Histopathology
- **Analysis:** Descriptive

RESULTS

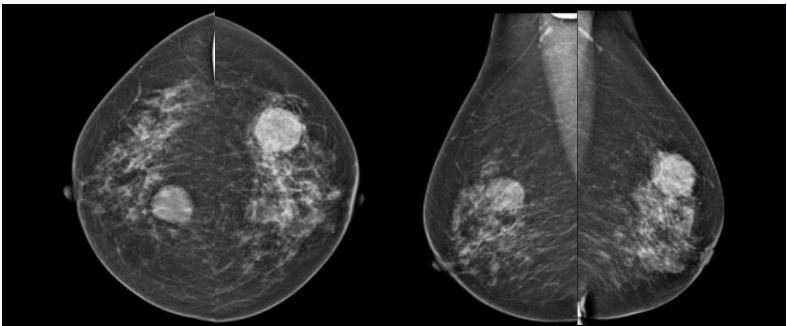


Figure 1. 2D Mammography

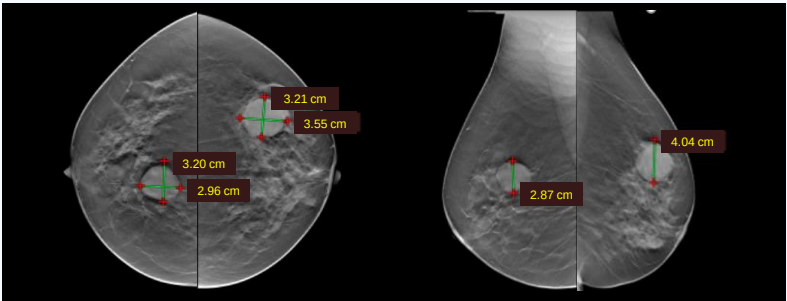


Figure 2. DBT (Digital Breast Tomosynthesis)

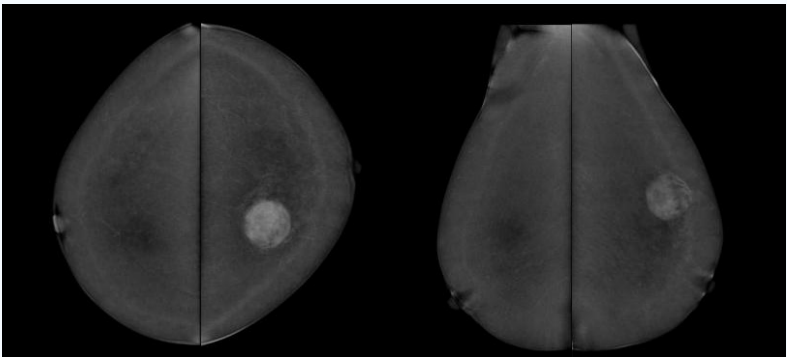


Figure 3. CEM (Contrast-Enhanced Mammography)

Key numerical findings

Left Breast (1 o'clock, Superolateral)

- **Mass:** Round, circumscribed, high-density mass, size 3.21 x 3.55 x 4.04 cm
- No skin thickening, nipple retraction, or bilateral axillary lymphadenopathy
- **CEM:** Intense enhancement
- **BI-RADS:** 4

Histopathology (Left)

- **Borderline phyllodes tumor**, size 55 mm
- Mild-moderate stromal atypia, increased stromal cellularity, **stromal overgrowth**, **3 mitoses/10 HPF**
- **Ki-67:** Positive in ~10% of tumor cells

Right Breast (11-12 o'clock, Superior)

- **Mass:** Round, circumscribed, equal-density mass, size 3.20 x 2.96 x 2.87 cm
- No skin thickening, nipple retraction, or bilateral axillary lymphadenopathy
- **CEM:** No pathological enhancement
- **BI-RADS:** 3

Histopathology (Right)

- **Fibroadenoma** with fibrocystic changes
- Usual ductal hyperplasia
- **Intraductal papilloma**

CONCLUSION / TAKE-HOME MESSAGE



1. Main finding

CEM demonstrated intense enhancement in a borderline phyllodes tumor and no pathological enhancement in fibroadenoma.



2. Clinical meaning

CEM complements DBT by providing functional information that may help differentiate similar fibroepithelial lesions.



3. Next step

CEM can guide tissue diagnosis and support appropriate clinical management and follow-up planning.

REFERENCES

1. Tan PH, Jayabaskar T, Chuah KL, et al. Phyllodes tumors of the breast: the role of imaging in diagnosis and management. Radiographics. 2016;36(3):707-723.
2. Jochelson MS, Dershaw DD, Sung JS, et al. Contrast-enhanced mammography. Radiology. 2013;266(1):43-49.
3. WHO Classification of Tumours Editorial Board. Breast Tumours. 5th ed. IARC;2019.
4. ACR BI-RADS Atlas. Breast Imaging Reporting and Data System. 6th ed. American College of Radiology; 2026.
5. Newell MS, Destounis SV, Leung JWT, DeMartini WB, Lee CH, Eby PR. ACR BI-RADS® v2025 Manual. Reston, VA: American College of Radiology; 2025.
6. Lee CH, Phillips J, Sung JS, Lewin JM, Newell MS. Contrast Enhanced Mammography. In: ACR BI-RADS® v2025 Manual. Reston, VA: American College of Radiology; 2025

ACKNOWLEDGEMENTS

